

# Accuracy Survives, Calibration Doesn’t: Post-Training Quantization of Time-Series Foundation Models

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## Abstract

Time-series foundation models (TSFMs) deliver strong zero-shot probabilistic forecasts, but little is known about how they respond to post-training quantization (PTQ). We study 39 weight-quantization configurations spanning seven method families (RTN, torchao, bitsandbytes INT8/NF4, HQQ, GPTQ, GPTAQ, AWQ-style scale folding) at two to eight bits, zero-shot on the 27-dataset Chronos benchmark across four models and three architectures (encoder-only Chronos-2, encoder-decoder Chronos-Bolt, decoder-only TimesFM-2.5), adding two calibration-aware diagnostics because a probabilistic forecaster can degrade in ways aggregate accuracy cannot see. Our central result is a *mechanism*: interval calibration and quantile monotonicity degrade *before* accuracy does, and in Chronos-2 the 3-linear quantile head—3.06% of the quantized weights—accounts for essentially all of that damage; an fp32 head restores fp32-level calibration at both bit-widths we test (W8, W4). Around it: 8-bit is practically free (all eight W8 arms within 1.3% WQL), yet one per-tensor 8-bit arm loses four points of coverage while looking lossless on WQL; error-compensating PTQ stays near parity at 3 bits (GPTAQ 1.016 [0.988, 1.053]) and every method collapses at 2; and method matters more than bit-width (0.996–1.66 at 4 bits). Post-hoc sorting and approximate recalibration remove crossing and restore nominal coverage—but recalibration repairs a model’s *claims* about its uncertainty, not its forecasts.

## 1. Introduction

Time-series foundation models (TSFMs)—Chronos (Ansari et al., 2024), Chronos-2 (Ansari et al., 2025), TimesFM,

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Moirai—forecast unseen series zero-shot and are increasingly deployed where full-precision inference is costly: CPU-only servers, edge devices, high-throughput batch pipelines. Quantization is the standard efficiency lever for large language models (LLMs), but its lessons do not obviously transfer to TSFMs, which are small (Chronos-2 has 120M parameters, a regime where quantization bites harder (Li et al., 2024)), non-autoregressive (no KV cache, so LLM decode-centric methods do not apply), and—critically—emit *probabilistic* forecasts whose calibration and quantile monotonicity matter, not just point accuracy.

**Relation to concurrent work.** Weight quantization of TSFMs has only just begun to be studied, and its probabilistic side is untouched. Concurrent work (TQS-PTQ; Pavlova et al., 2026) weight-quantizes TimesFM-2.5—our third architecture—plus Aurora-small and Pangu-Weather at W2–W4 against RTN/GPTQ/GPTAQ/QEP, on the ETT/Exchange/Weather suite with point metrics only. Ours is, to our knowledge, the first PTQ study of a TSFM under the standard *zero-shot* protocol, the first for the Chronos family and encoder-only quantile-head architectures, and the first to evaluate—and repair—the *probabilistic* output. The studies converge on where damage lives: TQS-PTQ ranks TimesFM’s tokenizer and point head most sensitive, independent evidence for the I/O-boundary mechanism we isolate. The nearest calibration neighbour is a conformal benchmark over quantized and sparse LLMs (Tong et al., 2026), classification-only and diagnostic rather than corrective.

**Contributions.** (1) A systematic PTQ study of TSFMs under a zero-shot probabilistic protocol: 39 configurations, 7 method families, 5 bit-widths, 4 models, 3 architectures, 27 datasets, with bootstrap confidence intervals on every aggregate. (2) A calibration-aware evaluation protocol for compressed probabilistic forecasters—interval-coverage error and a quantile-crossing rate—revealing failure modes invisible to WQL/MASE. (3) A *mechanism*: module-family sensitivity and weight-space SNR diagnostics identify the 3-linear quantile head (3.06% of quantized weights) as the gate on calibration damage in Chronos-2. (4) Two mitigations with a deployment recipe—an fp32 head (prevention) and post-hoc sorting plus approximate recalibration (cure)—

with an explicit account of what recalibration does *not* fix.

## 2. Setup

**Models.** Chronos-2 (Ansari et al., 2025) (amazon/chronos-2) is a 119.5M-parameter, T5-derived *encoder-only* transformer (12 layers,  $d_{\text{model}}=768$ , RoPE, alternating “time” and “group” attention) emitting, in one non-autoregressive pass, direct multi-step forecasts at 21 quantile levels; it ships in fp32 (478 MB). Quantization targets all 126 nn.Linear weights—including the patch embeddings and the 3-linear quantile head (3,649,536 of 119,439,360 linear parameters, 3.06%)—except in arms that explicitly protect the head; LayerNorm stays in fp32 and bf16 is the deployment baseline. As cross-model arms we also quantize Chronos-Bolt small/base (Ansari et al., 2024) (48M/205M), *encoder-decoder* T5 forecasters (in fp32 our harness reproduces their published per-task WQL to 0.34%/0.72% and MASE to 0.61%/1.38%, small/base; head protected by default), and TimesFM-2.5 (google/timesfm-2.5-200m-pytorch; 231M parameters as loaded), a *decoder-only* forecaster with a continuous quantile head, via a torch-only adapter (head and tokenizer protected). Its built-in `fix_quantile_crossing` auto-sort is disabled in all arms so QCR is measured honestly.

**Methods.** We evaluate weight-only PTQ across seven families runnable on commodity hardware: **RTN** (symmetric round-to-nearest, per-channel/per-tensor, plus a simulated sub-8-bit sweep); **torchao** INT8 weight-only and INT8 dynamic-activation (W8A8); **bitsandbytes** LLM.int8() (Dettmers et al., 2022) and NF4 (Dettmers et al., 2023); **HQQ** (Badri & Shao, 2023), a calibration-free half-quadratic solver; **GPTQ** (Frantar et al., 2023), a Hessian-aware error-compensating quantizer with one-shot calibration; **GPTAQ** (Li et al., 2025), its asymmetric-calibration extension correcting each layer toward the *full-precision* network’s activations; and **AWQ-style** activation-aware scale folding (Lin et al., 2024). Methods are exercised at W8, W4 (group size 64/128), W3, and W2—39 full-benchmark configurations in total.

**Protocol & metrics.** We evaluate zero-shot on the 27-dataset Chronos benchmark (Ansari et al., 2024) (190,674 forecasts) with the `fev` library—the harness the Chronos-2 authors use—so our numbers are comparable with the public leaderboard. Point accuracy is MASE, probabilistic accuracy weighted quantile loss (WQL, 9 levels); our fp32 Chronos-2 attains 0.425 WQL skill over seasonal-naive, beating it on WQL on all 27 tasks and on MASE on 88.9%, and our seasonal-naive matches the published per-task MASE/WQL to  $<10^{-6}$ . The key quantity is **retention**: the geometric-mean ratio  $\text{WQL}(\text{quant})/\text{WQL}(\text{fp32})$  over the 27 tasks (1.00 = parity), with a 1000-resample

Table 1. Representative variants, full 27-task benchmark. Retention = geometric-mean  $\text{WQL}(\text{quant})/\text{WQL}(\text{fp32})$ , [95% bootstrap CI]; *c80* = coverage of the nominal-80% interval (fp32 0.750); *QCR* = quantile-crossing rate (fp32 0.039). BPW = effective bits/weight incl. scales; † accuracy-exact simulation—BPW is the *achievable* footprint. “+head” = all linears quantized except the 3-linear head; “pt” = per-tensor scales.

| Variant            | BPW   | WQL ret. [CI]              | MASE  | c80  | QCR  |
|--------------------|-------|----------------------------|-------|------|------|
| fp32 (ref)         | 32.0  | 1.000                      | 1.000 | .750 | .039 |
| RTN W8             | 8.03  | 0.999 [0.997,1.001]        | 1.001 | .749 | .149 |
| RTN W8 (pt)        | 8.00  | 1.007 [1.000,1.014]        | 1.011 | .709 | .256 |
| HQQ W8             | 9.00  | 1.002 [1.0001,1.004]       | 1.000 | .746 | .086 |
| <b>RTN W8+head</b> | 8.03  | <b>0.999 [0.998,1.001]</b> | 1.000 | .751 | .037 |
| <b>GPTQ W4</b>     | 4.13† | <b>1.002 [0.985,1.023]</b> | 1.011 | .756 | .254 |
| HQQ W4             | 5.00  | 1.060 [1.017,1.124]        | 1.027 | .630 | .420 |
| NF4 W4             | 4.13  | 1.065 [1.001,1.149]        | 1.026 | .686 | .396 |
| HQQ W4+head        | 5.00  | 1.020 [0.986,1.067]        | 1.016 | .751 | .021 |
| NF4 W4+head        | 4.13  | 1.045 [0.979,1.122]        | 1.025 | .761 | .022 |
| GPTQ W3            | 3.25† | 1.047 [0.988,1.137]        | 1.059 | .791 | .366 |
| <b>GPTAQ W3</b>    | 3.25† | <b>1.016 [0.987,1.050]</b> | 1.034 | .770 | .379 |
| HQQ W3             | 4.20  | 1.229 [1.091,1.481]        | 1.143 | .666 | .589 |
| RTN W2             | 2.0†  | 4.082 [3.083,5.566]        | 3.108 | .060 | .523 |

bootstrap 95% CI. Because a probabilistic forecaster can lose quality without moving WQL, we add **MACE**, the mean  $|\text{empirical} - \text{nominal}|$  coverage error over the central 80/60/40/20% intervals, and **QCR**, the fraction of forecast points with non-monotone predicted quantiles. Two further protocols: a *module-family sensitivity* sweep (dev tier: 8 tasks) partitioning the 126 linears into *five* disjoint families—patch embed (3), time attn (48), group attn (48), FFN (24), head (3)—quantized or protected one at a time, plus two positional controls (first/last block) that deliberately cut *across* families; and an offline *repair* arm applying quantile sorting and approximate 2-fold series-split recalibration to the persisted predictions of 22 full-tier Chronos-2 arms.

## 3. Results

Table 1 summarizes representative configurations; Fig. 1 plots the accuracy-bit curve and calibration. All numbers are sourced from one bootstrap over the full 27-task benchmark; the complete 39-variant matrix is released.

**F1—8-bit is practically, not literally, free.** All eight W8 configurations land within 1.3% of fp32 on WQL (0.999–1.013) and 1.1% on MASE, at  $4\times$  compression—practical equivalence at a 1.3% margin, not statistical parity: five of the eight (HQQ, torchao INT8-wo and W8A8, per-tensor RTN, bitsandbytes INT8) have bootstrap CIs whose lower bound reaches 1.000. Accuracy also hides calibration: the per-tensor RTN arm is within 0.7% on WQL yet drops c80 from 0.750 to 0.709 and lifts QCR to 0.256 (Table 1)—an 8-bit model that looks lossless and is not.

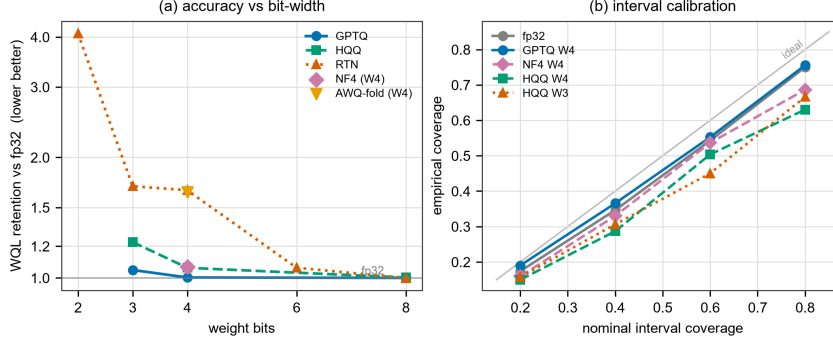


Figure 1. (a) WQL retention vs weight bits by method (log scale; lower is better, 1.0 = fp32). GPTQ tracks fp32 to 3 bits; the method spread at 4 bits exceeds the 8→4-bit gap of any single method; AWQ-fold sits on the RTN line. (b) Empirical vs nominal interval coverage: GPTQ-W4 tracks fp32 while HQQ/NF4 systematically under-cover.

**F2—error-compensating PTQ stays near parity at 3 bits; everything collapses at 2.** GPTQ is at parity at 4 bits (1.002, both group sizes). At 3 bits GPTAQ reaches 1.016 [0.988, 1.053]—the tightest sub-4-bit interval we obtain—and GPTQ 1.047 [0.988, 1.137]: we cannot reject parity for either, but GPTQ’s interval still admits  $\sim 14\%$  degradation, so GPTAQ *tightens* the W3 result rather than provably beating it (intervals overlap heavily; no paired test was run). At 3 bits HQQ loses 23% and naive RTN 70% (Fig. 1a), and GPTAQ’s asymmetry strength matters ( $\alpha=0.25$ : 1.016 vs  $\alpha=1.0$ : 1.087). At 2 bits every method collapses—GPTAQ 1.94, GPTQ 2.17, RTN-sim 4.08—so sub-3-bit deployment needs QAT.

**F3—method dominates bit-width.** At 4 bits, WQL retention spans 0.996 (GPTQ) to 1.66 (RTN), whereas moving a *good* method from 8 to 4 bits costs  $\leq 6\%$  (Fig. 1a): *how* you quantize matters far more than *how much*, inverting the intuition that bit-width is the primary knob.

**F4—calibration degrades before accuracy, and the quantile head gates it.** At 8 bits accuracy is retained yet the quantile-crossing rate already quadruples (0.039 → 0.149 for RTN)—a defect WQL and MASE cannot see; at 4 bits HQQ and NF4 sag to c80 0.630/0.686—a nominal “80%” band holding 63%—while GPTQ keeps 0.756. Sensitivity experiments locate the cause; *numbers marked (d) are dev-tier* (8 tasks, W4 RTN-sim). Quantizing *only* the 3-linear head—3.06% of weights—costs WQL 2.115 [1.436, 3.116] (d), indistinguishable from quantizing all 126 linears (2.130 [1.455, 3.239] (d)); protecting *only* the head recovers to 1.143 [0.957, 1.420] (d) with *better-than-fp32* calibration (c80 0.778, QCR 0.002). The dev tier exaggerates absolute damage—all-quantized is 2.130 (d) but 1.656 full-tier for the same configuration—so that comparison holds within-tier only. Within it the head accounts for 87% of the excess WQL and 89% of the excess MASE: at W4-RTN it gates accuracy too. What distinguishes calibration

is that head protection restores it for *every* method on the full benchmark, whereas accuracy retention still varies by method (Table 1, “+head”): an fp32 head returns W8-RTN’s QCR to 0.037 (fp32: 0.039) and repairs W4 coverage—HQQ 0.630 → 0.751 (WQL 1.020), NF4 0.686 → 0.761 (1.045)—for  $\approx 11$ –13 MB. Damage is not positional: only-group-attn 1.001 (d), only-FFN 1.086 (d), and the first/last-block controls change nothing (2.098/2.103 (d)). Prevention is verified at W8 and W4; head-protected W3/W2 arms are not yet run. Accuracy-only evaluation thus overstates compressed forecaster quality (Dettmers et al., 2024).

**F5—damage is tail-concentrated.** HQQ-W3 averages 1.23 but its worst task is  $8.2\times$  and 44% of tasks exceed 1.10; GPTQ-W4’s median per-task ratio is 0.998 with only 4% above 1.10. Report the distribution, not the mean.

**F6—no LLM-style outliers: the head fails on weight-space SNR, explaining the AWQ negative result.** The AWQ scale fold is *exact* for Chronos-2 (verified numerically), yet AWQ-fold+RTN at 4 bits (1.64) barely differs from plain RTN (1.66). Chronos-2 has *no* massive-magnitude activation outliers (global  $|x|_{\max} \approx 11$ ; at most 1.6% of channels exceed  $|x| > 6$ ), only ReLU-sparsity heterogeneity. The head instead fails on weight-space signal-to-noise: at W4 its output layer’s relative reconstruction error is 7.83, over  $100\times$  any other linear (all  $< 0.07$ ), because small quantile differences ride on large weights with no downstream layer to compensate. Activation *scaling* cannot fix a weight-space SNR deficit; Hessian-aware *compensation* (GPTQ/GPTAQ) can.

**F7—post-hoc repairs, and their limits.** Sorting predicted quantiles never increases pinball loss;<sup>1</sup> empirically it zeroes

<sup>1</sup>For fixed  $y$  and levels  $t_1 < \dots < t_K$ ,  $\partial_q \rho_t(y - q) = \mathbf{1}\{q > y\} - t$  is nondecreasing in  $q$  and decreasing in  $t$ , making the assignment cost submodular in  $(q, t)$ ; the sorted arrangement therefore minimizes  $\sum_k \rho_{t_k}(y - q_{(k)})$  pointwise, hence in any aggregate. Chernozhukov et al. (2010) instead prove rearrangement weakly

QCR for *every* arm at never-worse WQL and recovers accuracy where crossing is severe (HQQ-W3 1.229  $\rightarrow$  1.170). It is already standard practice—TimesFM ships an auto-sort on by default—so our contribution is the negative result that the *standard* sort suffices; the need is also general rather than quantization-specific (with that auto-sort disabled, TimesFM’s *fp32* model already crosses, QCR 0.109). Sorting only partly restores coverage (HQQ-W4 c80 0.630  $\rightarrow$  0.703); approximate per-level recalibration (2-fold series split, sort-before-calibrate) closes the rest, returning c80 to the 0.78–0.81 band for all 22 Chronos-2 arms, from RTN-W2-sim (0.060  $\rightarrow$  0.782) to AWQ-W3 (0.480  $\rightarrow$  0.795), with even raw *fp32* improving (MACE 0.059  $\rightarrow$  0.017). *But this repairs the model’s claims, not the model.* RTN-W2-sim ends nominally covered with WQL retention still 3.41 (from 4.08)—a forecaster 3.4 $\times$  worse than *fp32* wearing honest intervals, because they were widened; AWQ-W3 moves only 1.50  $\rightarrow$  1.41. Coverage must be read subject to sharpness (Gneiting et al., 2007), so we report WQL at every stage. The procedure is also approximate: it pools residuals across horizon steps within a series and applies a point-level  $\lceil (n+1)\tau \rceil$  order-statistic correction (per-level one-sided residual quantiles, not CQR), so the exchangeable unit is really the trajectory and no finite-sample split-conformal guarantee holds under within-series dependence—98 of 594 (model, task) cells get *worse* coverage afterwards, including *fp32* on *monash\_australian\_electricity* (0.675  $\rightarrow$  0.608). Horizon-stratified calibration (Stankevičiūtė et al., 2021) is the principled fix, and is future work.

**F8—head-protected quantization transfers across architectures.** On encoder–decoder Chronos-Bolt small/base (48M/205M), whose quantile head our harness protects by default, *every* method is at parity (small W4-HQQ 0.994, W3-GPTQ 0.986; base 0.9997–1.005). On decoder-only TimesFM-2.5 (231M), likewise head- and tokenizer-protected, W8-RTN 0.998, W4-GPTQ 0.999 [0.947, 1.030] and W4-HQQ 1.009 are at parity while W3-GPTQ degrades (1.101 [1.058, 1.152])—its cliff sits one bit higher than Chronos-2’s, a genuine cross-model difference. Calibration moves by method: RTN and HQQ leave Bolt coverage untouched (base c80 0.782  $\rightarrow$  0.782/0.783; small 0.773  $\rightarrow$  0.773/0.774), but the Bolt GPTQ arms shift it *up* 3–4 points into over-coverage (base 0.815/0.815 at W4/W3; small 0.811/0.817); TimesFM stays within 0.686–0.715 of its *fp32* 0.709. Since the head is protected in every arm here, these results establish that *head-protected* quantization retains accuracy and calibration across three architectures; they cannot test the mechanism itself, which is demonstrated only on Chronos-2. A head-unprotected arm on another family is the obvious next experiment.

reduces  $L^p$  distance to the *true* quantile curve; Fakoor et al. (2023), improvement of the weighted interval score.

**Efficiency.** Storage-real 4-bit models occupy  $\approx 60$  MB ( $8\times$  smaller) and cut peak batch-1 GPU memory 493 to 87 MB, but throughput is flat ( $\sim 120$  vs 110 series/s at batch 256): dequantize-then-GEMM buys footprint, not speed.

## 4. Discussion, Limitations, Conclusion

**Recommendation.** (a) At 8 bits use any toolkit, but keep the quantile head in *fp32* and prefer per-channel to per-tensor scales. (b) For 4–3 bits use GPTQ or GPTAQ; below 3 bits PTQ collapses—use QAT. (c) Always sort predicted quantiles post-hoc: free, provably never worse, zeroes crossing. (d) When calibrated intervals matter, recalibrate—but as an *offline diagnostic*: as evaluated it draws residuals from other series’ realized futures, so it is not an online wrapper, and it restores coverage without restoring forecast quality. Report calibration, sharpness, and crossing alongside WQL whenever a probabilistic model is compressed.

**Limitations.** We study univariate zero-shot tasks; covariate/multivariate tasks and QAT are future work. Module-family numbers are dev-tier (8 tasks), overstating absolute damage. The head mechanism is established on Chronos-2 only; head-protected W3/W2 arms and head-unprotected arms on Bolt/TimesFM are not yet run, and W2 is represented only by RTN-sim. The repair arm covers 22 Chronos-2 arms—no Bolt or TimesFM—and its recalibration is approximate: residuals are exchangeable at the trajectory level but treated pointwise, so no finite-sample guarantee holds. GPTQ/GPTAQ/AWQ arms are accuracy-exact simulations (realized speed needs packed kernels we do not implement), and latency is hardware-specific—retention ratios are the transferable claim. We run no head-to-head against TQS-PTQ (Pavlova et al., 2026), whose benchmark and metrics differ, and exclude KV-cache methods such as TurboQuant (Zandieh et al., 2025).

**Conclusion.** On accuracy, TSFMs quantize remarkably well across three architectures: 8-bit is practically free and error-compensating PTQ holds near-full-precision accuracy at 3–4 bits. The failure mode that does appear—calibration and monotonicity degrading before accuracy—is, in Chronos-2, gated by 3.06% of the weights, preventable at W8 and W4 with an *fp32* head and partly curable post hoc. But the cure is honest only about the model’s claims: it restores coverage, not quality.

**Reproducibility.** Code, configs, lockfile, benchmark definitions, per-task summaries and run metadata are released; the 1,692 per-series prediction parquets are available on request. Regenerate tables/figures with `analysis.campaign`, `quantization.sensitivity`, `analyze|report`, and `repair.analysis.py`, `aggregate.py`, `make-figures.py` under `scripts/`.

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